Digital Multimeter



Category: **Electrical & Electronic Test Instruments** Type: **Short SOP** Priority: **High**

29 labs

Covers these lab items: Digital multimeter; Multimeter; Multimeter, Clamp-meter; Multimeters; Multi-metre; Multi-tester; Voltmeter / DIGITAL MULTIMETER

What it is used for

A digital multimeter measures electrical quantities such as voltage, current, and resistance, depending on its design. Select the correct function and range before testing a circuit or component.

How to use

1. Inspect the meter, test leads, probes, and insulation for damage.
2. Confirm the meter and leads are rated for the circuit being tested.
3. Insert leads into the correct COM and measurement terminals.
4. Select the required function and range, and choose AC or DC correctly.
5. Connect probes in parallel for voltage or in series for current, following the meter instructions.
6. For resistance or continuity, isolate power and discharge capacitors before testing.
7. Read the display and record the measurement with its units.
8. Remove probes safely, switch off the meter, and store the leads.

⚠ Safety

- Never measure resistance on an energised circuit.
- Never place a meter set to current directly across a voltage source.
- Keep fingers behind probe guards and use suitable PPE.
- Do not use damaged leads or exceed the meter's voltage and current ratings.

Care & storage

- Keep the meter dry and clean, and store it in a protective case.
- Replace weak batteries and damaged leads, and verify calibration when required.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.